

# Arun Kumar

## Lead Data Scientist

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### Summary

Ambitious **Lead Data Scientist** with overall 10.8+ Years' experience in **Automotive, Manufacturing and Fintech** Domain. 4.5+ years' experience in building predictive modelling, data processing, data mining in **Machine Learning** and **Deep Learning** concepts to solve challenging business problems using **Time Series Forecasting, Regression, Classification, LSTM, Language Model** and created intuitive and comprehensive **Dashboards in Power BI** for the business for decision-making purposes. Looking for a challenging role to utilize my Programming, Database, BI and Leadership skills to expand my knowledge about new and emerging trends in the IT sector. Holds a **Bachelor's in Electronics and Communication Engineering**.

### Job Responsibility

- ❖ Skilled in developing applications with **Machine Learning, Deep Learning, Python, Power BI**.
- ❖ Built ML and DL models using **Time Series Analysis** to predict warranty claims, warranty costs, material failure frequency, pricing forecasts, and profit/loss within the Automotive domain. Additionally, applied the models to monitor CPU, memory, and disk utilization across servers in the Banking domain.
- ❖ Designed predictive models using **Regression** algorithms for client projects in the Automotive sector and **Classification** algorithms for projects in the Banking sector.
- ❖ Developing and deploying an **AI-driven medical chatbot** designed to assist psychology patients.
- ❖ Working on the development of an **AI-driven Resume Management Portal**, streamlining candidate detail extraction and aiding HR and recruiters.
- ❖ Created intuitive and business-focused **dashboards in Power BI** for the business for decision making purposes.
- ❖ Created the **Power BI Dashboard** for the Manufacturing domain like Sales Turnover, Inventory, Customer Outstanding, etc. and in Automotive domain client for Extended Warranty projects.
- ❖ Worked on connecting **ADLS Gen2, Oracle, MSSQL Databases** in Python, and **IBM DB2** servers using python directly on **Red Hat Linux** application servers.
- ❖ Delivered end-to-end **AI/ML and BI documentation**, presenting findings and insights to senior management.

### Technical Skill

| Skill                             | Technology worked                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Programming/Scripting             | Python                                                                                                                                                                                                                                                                                                                                                                                      |
| Tools/IDE                         | Visual Studio Code, PyCharm, PySpyder, Jupyter Notebook, Google Colab, ServiceNow, Power BI Desktop.                                                                                                                                                                                                                                                                                        |
| Machine learning                  | Machine Learning – <b>Supervised and Unsupervised Learning</b> : Linear Regression, Logistic Regression, Decision Trees, Random Forest, Ensemble Learning, SVM, Anomaly Detection, Time Series Analysis/Forecasting, KNN, K-Means Clustering, Dimensionality Reduction, Recommendation Engine, PCA, Naïve Bayes, Data Analysis, Pandas, NumPy, Scikit learn, Data processing and profiling. |
| Deep Learning                     | TF, ANN, CNN, RNN, Faster RCNN, OpenCV2                                                                                                                                                                                                                                                                                                                                                     |
| Gen AI                            | Language Model – BERT, Google PRO, Langchain                                                                                                                                                                                                                                                                                                                                                |
| Natural Language Processing (NLP) | Text pre-processing - Noise Removals, Word Normalization like Stemming, Lemmatization, etc.                                                                                                                                                                                                                                                                                                 |
| Data Visualization                | <b>Power BI</b> , Seaborn, Matplotlib, Dash and Plotly                                                                                                                                                                                                                                                                                                                                      |
| Repository                        | Git, Bitbucket                                                                                                                                                                                                                                                                                                                                                                              |
| Database                          | MSSQL, Synapse and Oracle DB - (C.R.U.D – Create, Read, Update and Delete). For Import/Exporting the data's into/from the Database through python and Power BI.                                                                                                                                                                                                                             |
| Project Methodology               | Warranty Cost and claim estimation using Time Series model and Material Rejections using Regression model for Automotive domain. Monitoring on Server and Capacity Utilizations using Machine Learning. Power BI Dashboard for decision making. AI based chatbot and Resume management, Agile Methodology.                                                                                  |
| Operating Systems                 | Windows, Red Hat Linux                                                                                                                                                                                                                                                                                                                                                                      |

**Professional Experience:**

**Project 1:**

|                   |                                        |            |         |
|-------------------|----------------------------------------|------------|---------|
| Project Name      | AI Medical Chatbot for Mental Wellness | Start Date | Mar 24  |
| Owner             | Healthcare                             | End Date   | Nov 24' |
| Project Locations | Hyderabad                              | Team Size  | 2       |

**Project Description:** In this Project, we are developing and deploying an AI-driven medical chatbot designed to assist psychology patients by providing empathetic responses, mental health tips, and dynamically generated follow-up questions. Utilized the **Microsoft Phi-2 language model** to understand and respond to user inputs, detecting emergency and irrelevant phrases to ensure appropriate responses. Implemented features to dynamically generate context-aware questions, enhancing user interaction and support. Integrating the chatbot for deployment on both local machines and Android devices, ensuring accessibility and ease of use for patients.

**Environment & technology used:**

- ❖ **Microsoft Phi-2 language model**, PyTorch, Python, VS Code, Jupyter Notebook, Android Studio.

**Contribution:**

- ❖ Conducted thorough requirement gathering and analysis sessions with stakeholders to understand the needs and expectations for the chatbot's functionalities and user interactions.
- ❖ Selected and fine-tuned the **Microsoft Phi-2 language model** to deliver empathetic, contextually appropriate responses tailored to the needs of psychology patients.
- ❖ Designed and developed algorithms to **detect emergency and irrelevant phrases** within user inputs, ensuring the chatbot could provide appropriate responses and escalate serious issues to emergency services if necessary.
- ❖ Implementing **advanced NLP techniques** to dynamically generate **context-aware follow-up questions**, improving the chatbot's ability to engage with users and **provide personalized support based on previous interactions**.

**Project 2:**

|                   |                          |            |          |
|-------------------|--------------------------|------------|----------|
| Project Name      | Resume Management Portal | Start Date | Aug 2024 |
| Owner             | Recruitment/HR Domain    | End Date   | -        |
| Project Locations | Coimbatore               | Team Size  | 6        |

**Project Description:** Developed a robust portal to automate the extraction of candidate details such as name, phone number, email, technical skills, location, and years of experience from uploaded resumes in various formats, including PDFs, DOCX, and scanned images. Once HR uploads resumes to the portal, the system processes them to extract candidate details and stores the information in a centralized database. The extracted details are then made available for HR to view and manage directly within the portal.

**Environment & technology used:**

- ❖ Google Pro, BERT uncased models, Python, Transformers, PyMuPDF, PyTesseract, OpenCV, MSSQL.

**Contribution:**

- ❖ Implementing end-to-end resume data extraction using **BERT uncased models** and regex-based pre-processing for structured data output.
- ❖ Automated parsing for various resume formats, including **PDFs, DOCX, and scanned image-based** resumes, using PyMuPDF, Tesseract OCR, and OpenCV.
- ❖ Designed a pipeline to standardize extracted data and upload it to the MSSQL database, ensuring accurate and efficient record management.
- ❖ Addressed challenges related to scanned resumes by **integrating OCR for text extraction** and enhancing detection accuracy.
- ❖ Developed a batch processing script to handle multiple resumes, updating the is\_Processed flag in the MSSQL database for seamless integration.
- ❖ Collaborated on testing and validating models to ensure a high level of accuracy in extracting key candidate information.

### Project 3:

|                   |                          |            |           |
|-------------------|--------------------------|------------|-----------|
| Project Name      | Extended Warranty Claims | Start Date | June 2022 |
| Owner             | Automotive Industry      | End Date   | Nov 2022  |
| Project Locations | Chennai                  | Team Size  | 3         |

**Project Description:** In this Project, we are predicting the Extended Warranty Claims by previous data using Machine Learning concept called **Time Series Analysis**. Once we gathered data from Synapse DB, we tried building ML **Time Series** models using **PyCaret, FBProphet, Croston, SARIMAX** and LSTM and finally we predicted using FBProphet which gave an accuracy more than 95% for Extended Warranty Claims and Cost on coming weeks/months on respective bike models. With EW Claims predictions model, customer has now the ability to visualize the forecast of EW claims for a required horizon and helped customer to plan warranty store manpower capacity to handle service load. It also helped to approximate outflow due to claims received. Client is able to compare vehicle models based on claims forecast. With that, model giving overall accuracy of >95%.

#### Environment & technology used:

- ❖ Machine learning models like PyCaret, FBProphet, Croston, SARIMAX and LSTM, Python, Pandas, NumPy, Scikit learn, Jupyter Notebook, Azure ML and ADF.

#### Contribution:

- ❖ Involved in requirement gathering and Architecture design of the project for machine learning implementation.
- ❖ Building baseline models for the requirements with necessary data preparation and bought the data into **stationary** by using **Moving Average, Exponential Smoothing** and **Differencing** model method.
- ❖ Built Machine Learning Algorithm for Time Series using **PyCaret, FBProphet, Croston, SARIMAX** and FBProphet gave an accuracy of 95% for Extended Warranty Claims on various bike material spare parts. Once we gathered the data, created Time Series algorithm and predicted data which will be stored in Azure and present to the clients through Power BI Visualization.

### Project 4:

|                   |                               |            |          |
|-------------------|-------------------------------|------------|----------|
| Project Name      | Failure Frequency Predictions | Start Date | Nov 2022 |
| Owner             | Automotive Industry           | End Date   | May 2023 |
| Project Locations | Chennai                       | Team Size  | 3        |

**Project Description:** In this RE Project, we are predicting the Extended Warranty Failure Frequency by previous data using Machine Learning concept called **Time Series Analysis**. Once we gathered data from Synapse DB, we tried building ML **Time Series** models using **PyCaret, FBProphet, Croston** and LSTM and finally we predicted using FBProphet for predicting which gave an accuracy more than 80% for Extended Warranty Failure Frequency on coming weeks/months on respective bike/models. With Failures predictions model, client has now the ability to visualize the forecast of no. of failures for each of the vehicle part. It helps customer to plan warranty store & manpower capacity to handle service load and also it helped to analyze part failures towards quality improvement. Client is now able to compare vehicle models & EW products based on part Failures. Model giving accuracy of >80%

#### Environment & technology used:

- ❖ Machine learning models like PyCaret, FBProphet, Croston, SARIMAX and LSTM, Python, Pandas, NumPy, Scikit learn, Jupyter Notebook, Azure ML and ADF.

#### Contribution:

- ❖ Involved in requirement gathering and Architecture design of the project for machine learning implementation.
- ❖ Building baseline models for the requirements with necessary data preparation and bought the data into **stationary** by using **Moving Average, Exponential Smoothing** and **Differencing** model method.
- ❖ Built Machine Learning Algorithm for Time Series using **PyCaret, FBProphet, Croston, SARIMAX** and FBProphet gave an accuracy of 80% for FF on various bike material spare parts. Once we gathered the data, created Time Series algorithm and predicted data which will be stored in ADB and present to the clients through Power BI Visualization.

### Project 5:

|                          |                                     |                   |          |
|--------------------------|-------------------------------------|-------------------|----------|
| <b>Project Name</b>      | <b>Pricing Forecast Predictions</b> | <b>Start Date</b> | May 2023 |
| <b>Owner</b>             | Automotive Industry                 | <b>End Date</b>   | Oct 2023 |
| <b>Project Locations</b> | Chennai                             | <b>Team Size</b>  | 4        |

**Project Description:** Deciding the price of extended warranty products depends on the cost of each of the automobile part covered under the warranty. The profitability of warranty business directly depends on the price of the warranty products sold to the customer. Our cost prediction model helped client to predict the cost of each of the 22,000 parts. Based on the predicted cost, client was able to price their warranty products in a profitable manner. An extraordinary solution was accomplished through extraction of Vehicle Parts Pricing data directly from SAP. In an innovative approach, the parts were meticulously categorized based on the patterns of increasing, decreasing, or stable prices. Leveraging the power of Holt Winters and Moving Average models, an exceptional generic model was constructed to predict the prices for each of these distinct categories. Model giving accuracy of 88-90%

#### **Environment & technology used:**

- ❖ Machine learning models like Holt Winters and Moving Average, FBProphet, Croston and LSTM, Python, Pandas, NumPy, Scikit learn, Jupyter Notebook, Azure ML and ADF.

#### **Contribution:**

- ❖ Involved in requirement gathering and Architecture design of the project for machine learning implementation.
- ❖ Building baseline models for the requirements with necessary data preparation and bought the data into **stationary** by using **Moving Average, Exponential Smoothing** and **Differencing** model method.
- ❖ Built Machine Learning Algorithm for Time Series using Holt Winters and Moving Average models gave an accuracy of 90% for PF on various bike material spare parts. Once we gathered the data, created Time Series algorithm and predicted data which will be stored in ADB and present to the clients through Power BI Visualization.

#### **Project 6:**

|                          |                                                       |                   |          |
|--------------------------|-------------------------------------------------------|-------------------|----------|
| <b>Project Name</b>      | <b>Fuel Tank Stencil - Image Classification Model</b> | <b>Start Date</b> | Sep 2023 |
| <b>Owner</b>             | Automotive Industry                                   | <b>End Date</b>   | Jan 24   |
| <b>Project Locations</b> | Chennai                                               | <b>Team Size</b>  | 5        |

#### **Project Description:**

Client prints fuel tank serial number, logo and other images on each of the fuel tanks manufactured. Due to system issues, Fuel tanks were getting duplicate serial numbers or non-uniform images getting printed on both sides of fuel tank. This caused them rework in printing of over 40% of the fuel tanks resulting in loss of time and money. We performed through object detection and image processing Deep learning model helped them to detect the duplication in fuel tank numbers even before it's printed.

#### **Environment & technology used:**

- ❖ Deep Learning algorithm, Python, TF, Tesseract-OCR, Pytesseract, Opencv-python, Pandas, Matplotlib, SQL.

#### **Contribution:**

- ❖ Involved in requirement gathering and Architecture design of the project for Deep Learning implementation.
- ❖ Data were collected using in real time prod environment were camera fixed with proper lighting.
- ❖ OpenCV-python used for image processing & developed in-built solution to identify the blurred/non-blurred images. Image pre-processing with Morphological Transformations, Image Blurring (Image Smoothing) and Image Transformation was used to extract required information from the image.
- ❖ With help of TF image classification (VGG-16) and object detection models are built. Information from the fuel tank extracted by the Tesseract OCR. QC time reduced from 40 secs to 5 secs.
- ❖ Finally, Model deployed in Azure Production environment.

#### **Project 7:**

|                          |                                   |                   |          |
|--------------------------|-----------------------------------|-------------------|----------|
| <b>Project Name</b>      | <b>Power BI Reports/Dashboard</b> | <b>Start Date</b> | Jan 2024 |
| <b>Owner</b>             | Manufacturing Industry            | <b>End Date</b>   | Jun 2024 |
| <b>Project Locations</b> | Coimbatore                        | <b>Team Size</b>  | 2        |

Project Description:

Developed intuitive Power BI dashboards for the Finance Team, encompassing critical reports including **Sales Turnover, Inventory, Customer Outstanding, Vendor Payables, Profitability Statement, Working Capital Statement, and Cash Flow statements.**

Environment & technology used:

- ❖ Power BI Desktop, MSSQL, FortiClient.

Contribution:

- ❖ Led **end-to-end project lifecycle** from requirement gathering through to deployment and support.
- ❖ **Designed architecture** and conducted detailed **documentation** for Power BI dashboard implementation.
- ❖ Extracted and processed data from **MSSQL**, ensuring accuracy through **data cleaning, merging, and validation.**
- ❖ Developed complex **DAX formulas** to derive meaningful insights from raw data.
- ❖ Created **comprehensive dashboards** spanning multiple pages, enhancing user accessibility and decision-making capabilities.
- ❖ Collaborated with stakeholders to **refine dashboard** requirements and ensure alignment with business objectives.
- ❖ **Successfully published and maintained dashboards,** providing ongoing support and updates as needed.

Organization:

| Organization                  | Role/Designation    | Date of Joining | Duration |
|-------------------------------|---------------------|-----------------|----------|
| FIS Global Business Solutions | Data Scientist      | Jan 2014        | 8 Years  |
| SkyBridge Infotech            | Lead Data Scientist | Feb 2022        | Present  |

Thank You!!